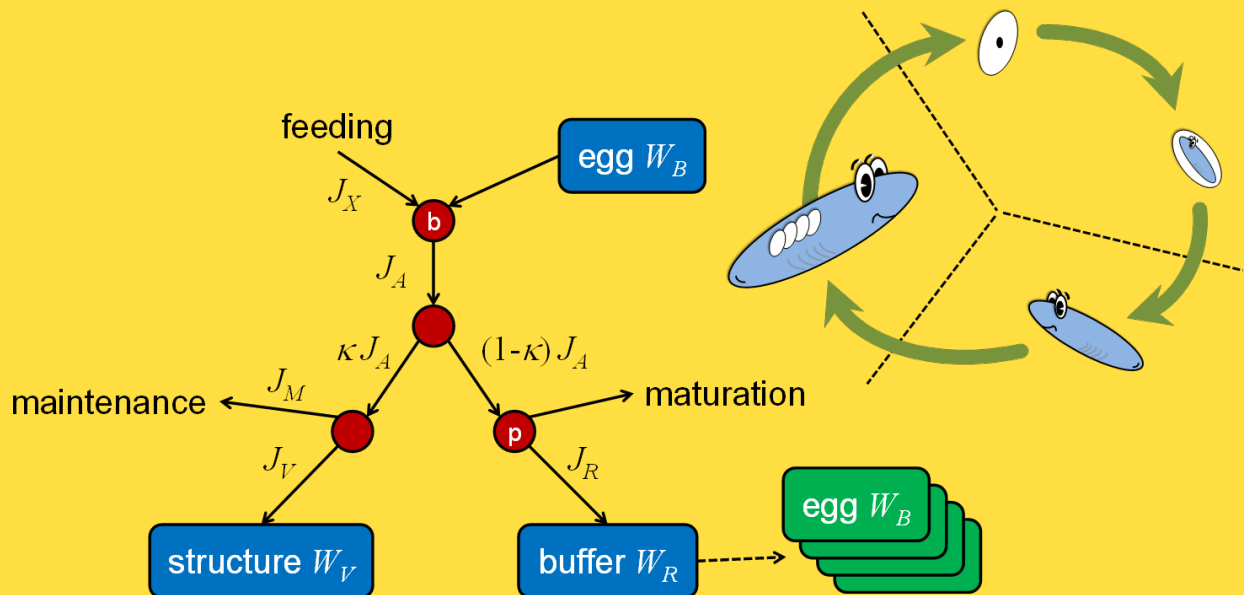


DEBkiss

A SIMPLE FRAMEWORK FOR
ANIMAL ENERGY BUDGETS



Tjalling JAGER

DEBkiss

A simple framework for animal energy budgets

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This e-book is published with Leanpub, and can be downloaded from https://leanpub.com/debkiss_book. The book can be obtained totally free of charge. However, please consider paying for it: that gives me the opportunity to continue working on this book, and on other ones. A version log can be found on http://www.debtox.info/book_debkiss.html. This e-book will (hopefully) be regularly updated and extended with new modules and case studies.

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Alternatively, refer to one of the papers in the open literature: e.g., [67] for the general DEBkiss model, and [50] for the TKTD module and the simplified ‘DEBtox’ model.

If you spot errors (spelling, grammar, mathematical or conceptual), please notify me by email (see address below). Furthermore, I love to receive feedback on this book, whether it has been readable and useful to you, and whether you are missing a topic. So feel free to drop me a line!

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Preface

The origin of DEBkiss

In 2002, I started working on the classical DEBtox model [93, 78]. I soon found out that the parameter governing the reserve dynamics (g) is almost impossible to obtain from data. Therefore, it was routinely fixed to a not-unreasonable value in practical applications. I never gave this much thought until, in 2011, Ben Martin showed me the problem in detail (as explained in his [113]), and noted that a (very detailed) data set suggests a very low reserve density for the water flea *Daphnia magna*. The discussion with Ben came back to my mind when I was preparing exercises for the 2012 summercourse on TKTD modelling (see <http://www.debttox.info/dynmodtox.html>). The standard DEB animal model [147] is, for my taste, too complicated as a place to start for such an introductory course, and the classical DEBtox formulation [93, 78] is not transparent enough and lacks the explicit mass balance (and is thus not the best choice, didactically). I then realised that ‘reserveless’ might well be the way to go: it often describes the data well and it is *much* easier to derive, interpret and apply. Together with a non-scaled explicit mass balance, this became DEBkiss [67]. At this moment, DEBkiss is my main tool for analysing and predicting (stress effects on) life-history traits, and so far I have not been disappointed by its ability to capture the life cycle of invertebrate animals (and even for vertebrates, in some cases).

In retrospect, I can now trace the seeds for this development even further back. In 2008/2009, I was preparing a paper with Chris Klok in which we compared three DEB models and how they differ in their predictions for population effects [64]. Chris had worked a lot with the ‘Kooijman-Metz’ model from 1984 [96], which is conceptually very similar to DEBkiss. I had just decided to continue working with the standard DEB model (due to the limitations of DEBtox), so I was not too eager to include such a ‘primitive’ model. However, I was struck by the performance of this simple model without reserve and maturity. At the time, I think I dismissed the good fits as a fluke, and something specific for the particular data set that we used.

About this book

The basis of this book is formed by a reshuffled and extended version of our paper in the Journal of Theoretical Biology [67], and therefore I like to acknowledge the contributions of my co-authors from that paper: Elke Zimmer and Ben Martin. This book combines the paper and its supporting information in a slightly different format, and I will add sub-models or modules to this book as time goes on. I also would like to thank Natnael Hamda and Alpar Barsi who were the first to apply DEBkiss in their PhD projects; I learned a lot from your studies. Finally, I wish to thank Chris Klok for (re-)introducing me to the Kooijman-Metz model.

If you are new to energy budgets, and want some more conceptual background, I strongly advise you to start by reading in my e-book “Making sense of chemical stress”, which can be downloaded from https://leanpub.com/debtox_book. That book provides a highly-readable and math-free introduction into energy-budget modelling (and its application in ecotoxicology), whereas I here immediately dive in and include the math (all pretty basic stuff though). The first three chapters form the backbone of this book: they explain and demonstrate the basic DEBkiss model. The remaining chapters are more technical, and more fragmentary, as they present a large number of potential model extensions with limited background information. I expect people to shop from these sections, if the need arises for their model application, rather than read it all.

At this moment, this book does not contain information about the statistical framework to use for model fitting to data, and error propagation. I might add this in a future version. However, in the meantime, I can refer you to the “Mechanistic modelling essentials” e-book (https://leanpub.com/mechmod_book), the supporting information of [78], and the more recent paper on the openGUTS parameter-space algorithm [51]. This algorithm has been included into the BYOM platform for more general use (available from <https://www.debtox.info/byom.html>). The technical background document for the “Making sense” e-book at <http://www.debtox.info/book.html> is now really outdated, but the information on likelihood functions is still valid.

Support on the web

Matlab packages are available in the BYOM library to play with DEBkiss: <http://www.debtox.info/byom.html>. There is a general package (DEBkiss) and a package DEBkiss_paper to reconstruct the case study of Chapter 3. However, the equations are simple enough to implement into the software of your choice. Further, there are two TKTD packages (DEBtox2019 and DEBkiss_tox) that are slightly more complex, and an experimental mixture-toxicity package DEBtox_mix. Several applications of DEBkiss have been (and are being) published, which I will try to collect at http://www.debtox.info/debkiss_appl.html.

Warnings

I noticed that some of my e-books are offered through other websites, after removal of the title page. Please download this e-book from Leanpub, which is the only place where it is legitimately offered (even for free). That will ensure that you have a complete, correct and safe version, and that you are notified as soon as a new version is available.

This book *will* inevitably contain errors. If you spot one, please let me know so that I can include corrections in updates of this book. I do not accept liability or responsibility for any damage or costs incurred as a result of these errors.

Chapter 1

Introduction

Simple is beautiful, but also practical, as embodied in the engineering principle of KISS (keep it simple, stupid). Complex things tend to break, and when they do, they are difficult to repair. But, the quote often attributed to Albert Einstein warns us not to overdo it: “everything should be made as simple as possible, but not simpler.” Here, we are going to apply the KISS principle to modelling of life-history traits of an animal (in particular: growth, development and reproduction), while heeding Einstein’s caution. How simple can we make a model for such traits of an individual, while still maintaining a degree of realism? This is one stage in a continuous quest for balancing simplicity and realism; a balancing act that will obviously depend on the purpose for which the model will be used. The specific purpose that I have in mind is to apply such a model for individuals of a species to interpret the effects of stressors such as toxicants [62] or food limitation [158], and to translate the effects on the individual to the population level [114, 64]. The focus in this work is on small invertebrate animals, although the models presented might well be useful for a broader range of organisms.¹

At minimum, our model should provide a prediction of reproductive output over the life cycle of an animal, as a function of food availability (which might vary over time). Reproductive output is the most straightforward indicator of individual fitness, and clearly needed for the translation to the population level; in its simplest form we can think of population dynamics as the difference between births and deaths. However, the reproduction rate is not determined by the current food level alone; it also depends on the state of the individual. Body size is an obvious candidate for such a state, as it determines feeding rates (and thereby the available resources for reproduction), and is often an accurate indicator of whether the organism is capable of reproducing. Interpreting the effects of varying food levels and stressors on reproduction therefore requires following (at least) body size as a state variable. Furthermore, because the dynamics of populations often depend on feedbacks between a population and its prey, keeping track of body size (and the associated feeding rates) is an essential aspect in population models.

Our model should thus provide us with a good description of at least body size and reproductive output over the entire life cycle (including the embryonic stage) as a function of food availability. It should be based on well-established principles (such as conservation of mass and energy, and consistency with thermodynamics), to ensure

¹Some applications with this model, and closely related ones, have been published for vertebrates in recent years. Especially for fish [68, 29, 112], but also application to mammals: mink [24] and rats [116, 115]. So, the application domain may be larger than I initially thought.

that the model behaviour is physically realistic. Furthermore, the model should include a (possibly crude) representation of biological processes such that we can model stressor effects on these processes. And finally, the core model should be generic and free from species- or stressor-specific argumentation as we do not want to build a new model for each species-stressor combination.

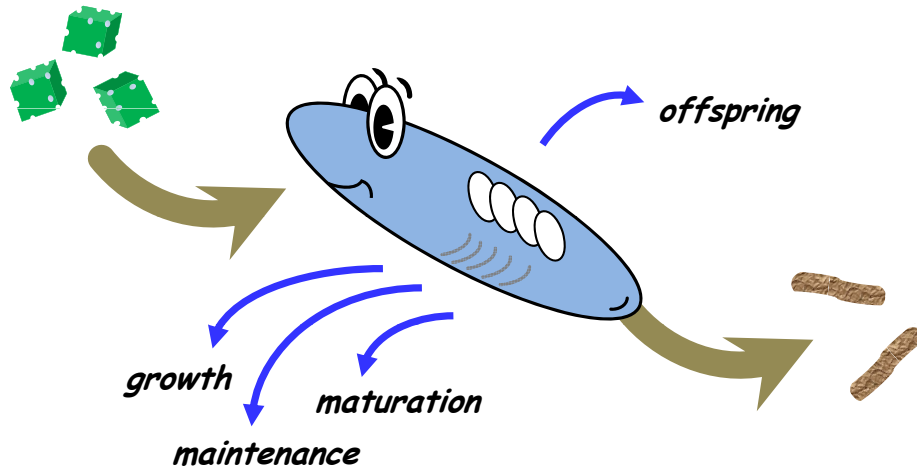


Figure 1.1: The energy budget is key to understanding energy-demanding traits such as growth and reproduction. Such traits are fuelled from the resources obtained by feeding, and thereby constrained by the conservation laws for mass and energy.

The best (and my opinion only) way to understand resource-requiring traits such as growth and reproduction is by focussing on the conservation laws for mass and energy. Growth and offspring production require energy, and compete with other energy-demanding processes such as maintenance of the body. This energy is ultimately obtained from feeding (in animals). Therefore, we need rules for how much food is taken up, and how its resources are allocated over the various energy-demanding traits (Fig. 1.1). Dynamic Energy Budget (DEB) theory is by far the most complete and most ambitious framework in this respect. DEB theory offers a powerful and formalised framework for building such models [86, 147, 129]. This power, however, comes at a price. Even though the concepts and underlying assumptions are mostly simple, understanding how they lead to the equations of the ‘standard DEB model’ for animals (see [147]) is not. Implementation of the model in software is certainly not straightforward, and the subsequent parametrisation requires an extensive data set (and even then it can be difficult to settle on a unique parameter set for a species, see e.g., [132]). Furthermore, the link between state variables and observable properties is rather awkward. Although efficient procedures and software have been developed to aid the user and to accommodate limited data sets [87, 102, 111], it takes some serious study to be able to apply them properly, and even more to verify the code. One would effectively have to rely on the derivations and programming of the developers, which can be an issue for potential users.

The standard model is the simplest complete DEB model, but it is often considered too complex (e.g., as a basis for population modelling, [128]). In many practical fields of application, the interest in dynamic models rapidly declines with the level of complexity. The standard animal model has been simplified, yielding the ‘scaled standard model’

[98, 77] and ‘DEBtox’ [78]. These simplifications, however, have their disadvantages. The use of scaling and compound parameters hampers interpretation of the equations and can lead to difficult-to-spot inconsistencies (e.g., transformation efficiency greater than one) for certain choices of parameter values.² Furthermore, the use of compound parameters hampers the straightforward application of stress due to toxicants, which are assumed to affect metabolic processes and thus primary energy-budget parameters [77].

In short, I believe there is room for a simple and transparent model that fully specifies the life cycle of an (invertebrate) animal, applies an explicit mass balance, and has rather direct access to the primary parameters and state variables that determine the metabolic processes. The model should be simple enough for users to check its consistency, implement into their own software of choice, and to parametrise it on easily-obtained data sets without additional help. Such a model would be suitable for particular applications where simplicity and transparency are of key importance, but it also provides a good teaching tool for theoretical biology in general, and will smooth entry into DEB theory in particular. In this book, I present such a simple model in a formalised manner (starting from an explicit formulation of the simplifying assumptions). I name the model ‘DEBkiss’ to emphasise that the work is highly inspired by DEB theory, but with a strong focus on the KISS principle.

²The original derivation of Kooijman & Bedaux [93] contained a few errors. Identified and corrected later [18, 78].

Chapter 2

Theoretical: the core model

2.1 Symbols and notation

The symbols for the basic DEBkiss model, with their dimensions, are explained in Table 2.1 (more symbols will be added in subsequent chapters). The first section of the table shows the primary parameters: parameters that are directly linked to a metabolic process, and that do not themselves depend on other parameters. In contrast, the values of secondary or compound parameters (bottom of the table) are fully determined by one or more primary parameters. As an example, maximum volumetric length L_m (the cubic root of maximum body volume) is a secondary parameter, whose value is determined by the primary parameters κ , J_{Am}^a and J_M^v (see Eq. 2.19).

It is always a struggle to come up with a notation that is fully internally consistent, easy to use, and intuitive (e.g., consistent with the expectations of the readers). The main problem is that these goals are mutually exclusive. Thus, the notation used here is a compromise. As much as possible, I tried to follow the DEB convention to use the same leading symbol for parameters with the same dimensions. However, this turned out to be impractical, and led to too many subscripts to distinguish between different parameters. One of the solutions is to work with superscripts in addition to subscripts. This has some drawbacks, mainly that a superscript can be interpreted as ‘to the power’. Fortunately, the superscripts are only used on parameters that are not part of power functions. A minor drawback is that word processors such as MS Word have trouble with simultaneous sub- and superscripts in text mode.

The superscripts are used to indicate parameters that are volume- or surface-area-specific. As an example, J_M^v is the volume-specific costs for maintenance, and J_{Am}^a is the area-specific assimilation rate at maximum food. Therefore, the dimensions also differ from those of the fluxes J_i (which are in mass per time). Furthermore, superscripts are used when I discuss toxicants in Chapter 5. There, they are used in a different but related manner to indicate that a parameter is scaled or referenced in a certain manner. For example, D^w is used for a scaled damage level, where the damage is referenced to the external concentration in water. Hence, D^w has the dimensions of an external concentration. As another example, k_e^m is the elimination rate for a fully-grown animal; for smaller individuals, this reference value is scaled using the ratio of the actual body size with the maximum body size.

Symbol	Explanation	Dimension	Sugg. value
Primary parameters			
F_m^a	Maximum area-specific searching rate	$l_e^3/(l^2t)$	—
J_{Am}^a	Maximum area-specific assimilation rate	$m_a/(l^2t)$	—
J_M^v	Volume-specific maintenance costs	$m_a/(l^3t)$	—
W_{B0}	Assimilates in a single freshly-laid egg	m_a	—
W_{Vp}	Structural body mass at puberty	m	—
y_{AV}	Yield of assimilates on structure (starvation)	m_a/m	0.8 mg/mg (dwt)
y_{AX}	Yield of assimilates on food	m_a/m_f	0.8 mg/mg (dwt)
y_{BA}	Yield of egg buffer on assimilates	m_a/m_a	0.95 mg/mg (dwt)
y_{VA}	Yield of structure on assimilates (growth)	m/m_a	0.8 mg/mg (dwt)
κ	Fraction of assimilation flux for soma	—	0.8
Conversions			
d_V	Dry-weight density of structure	m/l^3	0.1-0.2 mg/mm ³
$\delta_{\mathcal{M}}$	Shape correction coefficient	—	
Fluxes, states and forcings			
J_A	Mass flux for assimilation	m_a/t	
J_M	Mass flux for maintenance	m_a/t	
J_R	Mass flux to reproduction buffer	m_a/t	
J_V	Mass flux for structure	m/t	
J_X	Mass flux of food	m_f/t	
W_B	Mass of assimilates buffer in egg	m_a	
W_R	Mass of reproduction buffer in adult	m_a	
W_V	Mass of structural body	m	
X	Food density in the environment	m_f/l_e^3	
Other output and secondary parameters			
f	Scaled functional response (0-1)	—	
J_{Xm}^a	Maximum area-specific feeding rate	$m_f/(l^2t)$	
X_K	Half-saturation food density	m_f/l_e^3	
L	Volumetric body length	l	
$L^{\mathcal{M}}$	Physical body length	l	
r_B	Von Bertalanffy growth rate constant	$1/t$	
ΔR	Number of eggs in a clutch	$\#$	
R	Continuous reproduction rate	$\#/t$	
R_m	Maximum continuous reproduction rate	$\#/t$	
t_b	Time between egg laying and birth	t	
W_{Vb}	Structural body mass at birth	m	
W_W	Total body weight	m	

Table 2.1: Explanation of symbols for the basic model, with dimensions given in mass (m for body, m_a for assimilates, and m_f for food), length (l_e for environment, l for organism), numbers ($\#$), time (t). Suggested values for the yields (apart from y_{AV}) based on the typical values in [102].

2.2 Model definition

The DEBkiss model is schematically depicted in Figure 2.1, showing the mass fluxes J_* (in dry weight per unit of time). In the possible topologies for energy budget models of [103], it would classify as a $R\kappa_G^{R0}$ model. In the κ models, the fundamental split between investment in the soma and reproduction comes first (on the assimilates obtained from feeding). This contrasts ‘production models’, where maintenance costs are paid before the split (e.g., [106]), and ‘assimilation models’ where the split comes after a storage compartment (e.g., [78, 147]). This topology maintains many of the desirable properties of the standard DEB model [103], especially for small animals, while considerably simplifying the model equations.¹

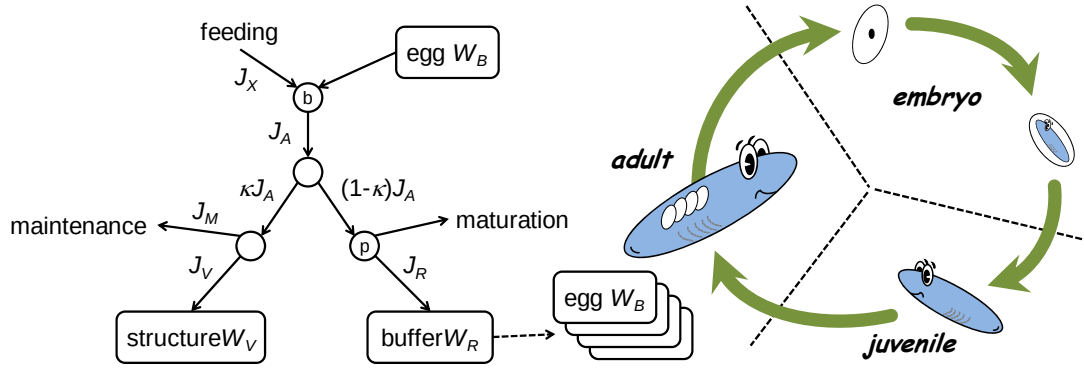


Figure 2.1: Schematic diagram of the energy flows and life cycle of a DEBkiss animal. The parameter symbols are explained in Table 2.1. The nodes b and p denote switches at birth (start of feeding; embryo to juvenile) and puberty (start of reproductive investment; juvenile to adult). The other nodes represent a split of the assimilation fluxes.

The model departs from a strict set of assumptions, which lead (uniquely) to the model equations. Because assumptions form the heart of modelling, they are placed in boxes for emphasis.

Assumptions 1: There are three types of biomass: food, assimilates and structural body components. Each type has a constant composition. They can be converted into each other with a certain constant efficiency. The state variables of the organism are the masses of the structural body, the reproduction buffer for adults, and the egg buffer used by the developing embryo. Total body mass is the sum of structure and reproduction buffer in adults, and the sum of structure and egg buffer for eggs. The reproduction and egg buffer consist of assimilates.

The ‘currency’ that we are going to follow in the model is mass as dry weight (e.g., in grammes). However, we can substitute mass for energy: because we assume that each type of biomass has a strictly constant composition, the conversions between mass and energy are also constant. The choice of currency does however have repercussions for the transformations, such as the yield of assimilates on food y_{AX} . Consider for example an earthworm feeding on soil. Even if the worm is able to extract the majority of the energy from its food, indigestible sand and clay particles form the bulk of the ingested soil. The y_{AX} based on energy will thus be much higher than when based on mass. The

¹See Appendix B for a detailed analysis of the empirical patterns that [103] use to compare topologies.

assumption of a constant composition for each type of biomass is made for practical reasons. This allows us to use constant conversions between each type, and means that we do not have to follow individual components of biomass such as lipids and proteins.

Assumptions 2: The animal has three life stages: an embryo that does not feed but utilises the egg buffer, a juvenile that feeds but does not reproduce, and an adult that feeds and reproduces. The embryo starts with an egg buffer of assimilates and negligible structural mass. The first transition (birth) is triggered by the depletion of the egg buffer, and the second transition (puberty) by reaching a critical structural body weight.

The differential equations for the egg buffer W_B , structural body mass W_V , and reproduction buffer W_R are given by (see Fig. 2.1):

$$\frac{d}{dt}W_B = -J_A \quad \text{until } W_B = 0, \text{ with } W_B(0) = W_{B0} \quad (2.1)$$

$$\frac{d}{dt}W_V = J_V \quad \text{with } W_V(0) \approx 0 \quad (2.2)$$

$$\frac{d}{dt}W_R = J_R \quad \text{with } W_R(0) = 0 \quad (2.3)$$

Note that $t = 0$ here marks the start of development in the egg.² The total weight of the animal is the sum of structure and any buffer that is present; for adults, this may require consideration of the reproduction buffer ($W_W = W_V + W_R$), and for the egg the egg buffer ($W_W = W_V + W_B$). For some processes, we need to have access to the volume (L^3) of the animal. As composition of each biomass component is assumed to be constant, it follows that we can use a constant density for structure (d_V):

$$L^3 = \frac{W_V}{d_V} \quad (2.4)$$

We can talk about L as the ‘volumetric structural length’ of the animal. If the structural biomass W_V is compressed into a cube, this will be the length of a side of that cube. The value for the density d_V will not generally influence the model fit on growth and reproduction data; it mainly influences the numerical value of the area- and volume-specific parameters, and is thus important to compare species. Furthermore, when including other endpoints such as ingestion or respiration rates, selecting a representative value for d_V becomes essential.³

Because egg weight (W_{B0}) is a primary parameter, it can vary independently from the other primary parameters. Therefore, we are free to vary the egg weight (or include descriptive functions) to match patterns that are observed in practice. For example, there are many species in which the investment per offspring increases with size or age of the mother [17, 30]. An unrealistic consequence of our assumption for the birth trigger ($W_B = 0$) is that the embryo will always hatch from the egg, even when its development is hampered by a stressor or when the egg buffer is experimentally reduced (e.g., extraction of yolk). In reality, the embryo will need to reach a certain minimum

²For some species, the start of the development may precede egg laying (e.g., *Caenorhabditis elegans*), or development may start well after egg laying (when there is a diapause).

³In [102], $d_V = 0.1$ is suggested as general default. However, there are substantial differences between taxa (see e.g., [80, 134] for marine groups). For arthropods, a value of $d_V = 0.2$ is probably a more realistic default.

amount of complexity, or a minimum body size, to be able to survive hatching.

Assumptions 3: The maximum feeding rate is proportional to the surface area of the animal. The animal is either searching for food or handling it (with constant handling time), leading to a hyperbolic functional response in the food density (Holling type II).

Feeding involves the transport of resources from the environment to the organism across a surface area (e.g., the area of the gut, or the area of the feeding appendages in filter feeders). As long as the organism does not change in shape (isomorphy), all surface areas scale with body volume to the power $2/3$ (and thus L^2). We can easily experiment with other scalings for the full life cycle, or just a part of it, which might be realistic for specific species (see e.g., [97, 8]). The feeding flux is given by:

$$J_X = f J_{Xm}^a L^2 \quad \text{with } J_{Xm}^a = \frac{J_{Am}^a}{y_{AX}} \quad (2.5)$$

where f is the scaled functional response, which is the actual feeding rate at a certain food level divided by the maximum feeding rate for its current size. The scaled response f is thus between 0 (no food) and 1 (*ad libitum* food). The maximum specific assimilation rate (J_{Am}^a) is used as the primary parameter; the specific feeding rate (J_{Xm}^a) is derived from it, using the yield of assimilates on food (y_{AX}). When we do not follow feeding explicitly, we can use f as a primary model parameter. Otherwise, the forcing function of the food density X enters the system. The scaled functional response f is a hyperbolic function of the food density, and the half-saturation food density X_K is calculated from the specific feeding rate and the specific searching rate F_m^a :

$$f = \frac{X}{X + X_K} \quad \text{with } X_K = J_{Xm}^a / F_m^a \quad (2.6)$$

Assumptions 4: Food is instantly translated into assimilates that are directly used to fuel metabolic processes. Embryos assimilate their egg buffer at the maximum rate for their structural size.

We do not consider the ‘details’ of digestion because this process plays at a time scale that is usually very short relative to the life cycle of the organism. The assimilates are directly used in metabolism, and therefore, we do not consider any storage other than the egg and reproduction buffer. The assimilation flux J_A is thus given by:

$$J_A = f J_{Am}^a L^2 \quad (\text{if } W_B > 0 \text{ then } f = 1) \quad (2.7)$$

Assumptions 5: The flow of assimilates is split into a constant fraction κ for maintenance and structural growth (the soma), and $1 - \kappa$ for maturation and reproduction. From the κ flow, maintenance costs are paid first. Only structural biomass requires maintenance, which is proportional to its volume. The remainder of this flow is used for growth (with certain efficiency).

A constant κ has convenient properties, which compare favourably to other possible allocation rules [103]. This simple rule ensures that growth and reproduction do not compete for resources. This lack of competition is evidenced in many invertebrates as the start of reproduction (a major flux of resources) is not accompanied by changes in

the growth curve, respiration or feeding rates (see [129]). A constant κ , together with the assumptions for assimilation and maintenance, leads to the commonly-observed von Bertalanffy growth curve in constant environments.

Maintenance is the, rather abstract, lump sum of all the processes needed to maintain the body's integrity. Assimilate buffers are assumed not to require maintenance, which is supported by the almost-complete lack of respiration in freshly-laid eggs. The flux for structural growth (J_V) can thus be specified as:

$$J_V = y_{VA}(\kappa J_A - J_M) \quad \text{with } J_M = J_M^v L^3 \quad (2.8)$$

where J_M^v is the volume-specific maintenance cost, and y_{VA} is the yield of structural biomass on assimilates.

Assumptions 6: For adults, the $1 - \kappa$ flow is used to fill the reproduction buffer. For embryos and juveniles, all of the assimilates in this flux are burnt to increase complexity of the organism. At spawning events, the contents of the reproduction buffer are converted into eggs. The part of the buffer that was insufficient to create a single egg remains in the buffer. Transformation of buffer to egg comes with a certain (generally high) efficiency.

Assuming that κ is constant over the life cycle requires an explanation of what happens to the $1 - \kappa$ flow before puberty. Following DEB theory, I assume that this flux is used for the maturation process (which in the DEB context is not associated with the build-up of biomass), which abruptly stops at puberty, when the flux is switched to the reproduction buffer. The flux into the reproduction buffer (J_R) can thus be specified as:

$$J_R = (1 - \kappa)J_A \quad (\text{if } W_V < W_{Vp} \text{ then } J_R = 0) \quad (2.9)$$

where W_{Vp} is the body size where investment in reproduction starts (puberty). The trigger for spawning is not specified here, as this is highly species-specific. For some species, a constant time interval between clutches of eggs may be appropriate, while for others, a critical mass of the reproduction buffer may be more realistic. For others, spawning may be triggered by environmental conditions such as temperature or pH. Spawning leads to a clutch of offspring ΔR , and a reset of the reproduction buffer W_R :

$$\Delta R = \text{floor} \left(\frac{y_{BA} W_R}{W_{B0}} \right) \quad (2.10)$$

$$W_R = W_R - \frac{\Delta R W_{B0}}{y_{BA}} \quad (2.11)$$

where y_{BA} is the yield for the conversion of reproduction buffer to eggs. The ‘floor’ function for the spawning events means rounding to the nearest integer less than the value between brackets. Without reproduction buffer, a continuous reproduction rate R can be calculated as:

$$R = \frac{y_{BA} J_R}{W_{B0}} \quad (2.12)$$

Assumptions 7: If feeding is insufficient to pay somatic maintenance costs, the organism first diverts energy from the $1-\kappa$ flux of assimilates and from the reproduction buffer. If that is insufficient, structure is converted into assimilates to pay maintenance.

We need assumptions to deal with the situation of starvation because our animal does not have a storage of assimilates (other than the reproduction buffer). Time-varying food levels are common in the field, and also stress effects on parameter values (e.g., toxicity) could induce starvation, even under *ad libitum* feeding conditions (see [50]).

The first stage of starvation occurs when the allocated flux to the soma is insufficient to pay maintenance ($\kappa J_A < J_M$), but the total assimilation flux is enough ($J_A > J_M$), or there is still something in the reproduction buffer ($W_R > 0$):

$$J_V = 0 \tag{2.13}$$

$$J_R = J_A - J_M \quad (\text{if } W_V < W_{Vp} \text{ then } J_R = 0) \tag{2.14}$$

For juveniles, this means that energy is diverted from the flux to maturation, as long as $J_A > J_M$ (maturation itself is not followed as a state variable). For adults, if there is a reproduction buffer, J_R becomes negative when $J_A < J_M$ as the buffer is used to cover the difference between assimilation flux and maintenance needs. In the second stage of starvation, the reproduction buffer is absent or empty ($W_R = 0$) and the total assimilation flux is insufficient to pay maintenance ($J_A \leq J_M$). This implies that the animal needs to shrink to cover its maintenance costs:

$$J_V = (J_A - J_M)/y_{AV} \tag{2.15}$$

$$J_R = 0 \tag{2.16}$$

where y_{AV} is the yield of assimilates (to pay maintenance) on structure. The maximum rates of feeding, assimilation and maintenance depend on structural size, so when the animal shrinks, these rates will decrease too. A decrease of the maximum possible feeding and assimilation rates on shrinking might not be realistic for all species. For *Daphnia* for example, model correspondence to recovery after starvation improved by relating the assimilation rate to the previously obtained maximum size, rather than actual body size [113].⁴

Clearly, shrinking under starvation cannot continue indefinitely. If situations of prolonged starvation are analysed, it makes sense to set a limit to shrinking, e.g., to a fraction of the maximum size that the individual has reached. Furthermore, it might be realistic for some species to stop spawning (the conversion of reproduction buffer to eggs) to enhance starvation resistance, and perhaps for others to convert structure into eggs ('suicide reproduction'). This set of starvation rules should be seen as a start for experimentation; different rules may be more applicable for a particular species.

2.3 Derived model results

In many cases, body size of an animal is determined as some length measure. Examples are the shell length of snails and mussels, and the distance from the eye to the base

⁴A possible explanation is that the feeding appendages are part of the carapace, and hence do not shrink when structure is burned.

of the spine in daphnids. As long as the organism does not change in shape during growth, we can translate structural weight to some physical length ($L^{\mathcal{M}}$) and vice versa using a correction factor $\delta_{\mathcal{M}}$:

$$L^{\mathcal{M}} = \frac{L}{\delta_{\mathcal{M}}} \quad \text{where } L^3 = \frac{W_V}{d_V} \quad (2.17)$$

For a well chosen length measure, $\delta_{\mathcal{M}}$ can remain constant as the animal grows. However, special care must be taken when the animal is shrinking under starvation. If the length measure is based on a fixed structure of the animal, such as a shell or a carapace, the dry weight will decrease without an associated decrease in physical length (see [11]).

The DEBKiss model system explicitly includes the food density X . This is particularly important for including fluctuating food concentrations, and the influence of feeding on the food density. We may, however, wish to analyse results from laboratory tests at constant or *ad libitum* food levels. In that case, the scaled functional response f can be used as a primary model parameter. For *ad libitum* feeding, we can set $f = 1$, and estimate a value for f for constant limiting food levels (without the need to know the value of X). As a next simplification, we can lump several of the more abstract model parameters into compound ones. By filling in the fluxes (see Appendix A.1), and moving from structural mass to volumetric length L , it can easily be derived that the growth of the organism (after birth) will follow the von Bertalanffy equation:

$$\frac{d}{dt}L = r_B(fL_m - L) \quad \text{with } L(0) > L_b \quad (2.18)$$

Where L_b is the volumetric length at birth. The growth equation also applies to embryos, but for them, we set $f = 1$, as long as egg buffer is present. The two compound parameters for maximum volumetric length L_m and the growth rate constant r_B are linked to the underlying primary parameters:

$$L_m = \kappa \frac{J_{Am}^a}{J_M^v} \quad \text{and} \quad r_B = \frac{y_{VA}}{3d_V} J_M^v \quad (2.19)$$

We can thus obtain the maximum structural weight as $W_{Vm} = d_V L_m^3$. Note that the growth equation above does not require time to start at birth; we can take $t = 0$ anywhere after birth. In the absence of a reproduction buffer, the state variable W_R can be removed, and we can similarly fill in the fluxes for the continuous reproduction rate to obtain the maximum rate (at $f = 1$, $L = L_m$, and in the absence of stressors):

$$R_m = (1 - \kappa) J_{Am}^a L_m^2 \frac{y_{BA}}{W_{B0}} \quad (2.20)$$

When food density is constant or *ad libitum*, in the absence of stress effects on the parameters, and ignoring the reproduction buffer, the ODE for body size can be solved analytically. This yields a surprisingly compact model system (see [96, 82]):

$$L = fL_m - (fL_m - L_0)e^{-r_B t} \quad (2.21)$$

$$R = fR_m \frac{L^2}{L_m^2} \quad \text{when } L > L_p \quad (2.22)$$

This model is so simple it can easily be implemented into a spreadsheet, and contains only six model parameters (f , L_0 , L_p , L_m , r_B and R_m). Note that the parameter L_p is linked to W_{Vp} with the density d_V (see Eq. 2.17). However, there are limitations to take into consideration for this simplification. The parameters in the growth equation need to be constant, although we can always resort to using the ODE of Eq. 2.18. The essence of the model, the mass balance, is not obvious from these equations, which might give the impression that this is just some magical equation to describe the data patterns. Linked to this, it is possible to come up with parameter combinations that violate the underlying mass balance (e.g., yield coefficients larger than one). The reproduction buffer is lost, leading to a continuous production of offspring. Although we can easily modify R to become discontinuous at spawning events, we cannot include the reproduction buffer into the total body weight or use the buffer contents on starvation, because the mass of the eggs has been absorbed in R_m . And finally, stress should affect metabolic processes, and thus primary model parameters. Implementing stress on the compound parameters should be considered very carefully indeed (see also [78]).

If we can ignore the maintenance flux for embryos ($J_M \approx 0$), this allows for a simple explicit equation for birth weight and development time (and thus remove the state variable W_B from the system):

$$W_{Vb} = W_{B0} y_{VA} \kappa \quad (2.23)$$

$$t_b = \frac{3W_{B0}^{1/3} d_V^{2/3}}{J_{Am}^a (y_{VA} \kappa)^{2/3}} \quad (2.24)$$

This provides a good approximation of these two model outputs. In detail, the underlying assumption is unrealistic, but the influence of maintenance on the growth curve of embryos can be ignored in many cases as the structural body mass will be very small.

2.4 Extensions

The DEBKiss model offers an excellent platform for experimentation. After the general case study of Chapter 3, I work out a range of possible extensions that can be explored further in Chapter 4. An obvious extension would be to include maturity maintenance; a similar process to somatic maintenance, but applied to maturity in the $1 - \kappa$ branch in Figure 2.1. This process (worked out in Section 4.1) helps to explain why an individual at a low constant food level might remain in the juvenile stage indefinitely, despite continuous allocation to the $1 - \kappa$ branch. All other parameters being equal, adding this process leads to less reproductive output and a slightly different shape of reproduction rate versus body size. Even though I excluded this process from the bare-bones model description above, I tend to include it by default when using the model in practice (e.g., for ecotoxicological applications as in Chapters 5 and 6).

Some organisms seem to deviate from the expected von Bertalanffy growth curve under constant conditions, revealing a more S-shaped pattern (when body size is expressed as a length measure). For some species, there are strong indications that this is caused by an inappropriate food source for the juveniles in experimental tests [55, 158]. For others, a temporary acceleration of metabolism after birth seems appropriate [97, 8]. These options are worked out in Section 4.2. Additionally, extensions are presented to calculate respiration rates (Section 4.3), plus a number of bits and bobs (Section 4.4).

The Chapters 5 and 6 deal with the extension of the general model to stressor effects. For ecotoxicological applications, we need to add the uptake and elimination of chemicals (toxicokinetics) to the model, as well as the formation of ‘damage’, and from that to effects of chemicals on metabolic processes (toxicodynamics). The internal concentration in the individual can be linked to any of the primary parameters of the model [78, 77]. Toxic effects on survival can similarly be linked to the internal concentration, e.g., by using any of the toxicodynamic modules of the GUTS framework [54, 57].

In its current form, the DEBkiss model deals with growth, development and reproduction, but another important life-history process, ageing, is not treated. It is possible to simply add mortality as a descriptive function of age, but that ignores the link to metabolism (e.g., an effect of food level on longevity is commonly observed). The model offers sufficient possibilities to play around with process-based ageing modules, linked to the various metabolic processes (see [153]), but the most appropriate model in this context requires further study. A promising simple sub-model is proposed in Section 5.6, though it still awaits serious testing. This topic is treated in the chapter on toxic effects as it considers ageing to lead to damage in the organism, which affects life-history traits in a similar way as toxicants. The ecological relevance of an ageing module may be limited, however, as outside of the laboratory, animals do not generally have the luxury to die of old age.

2.5 Data requirements

Establishing model parameters requires information on the life history of a species. It is difficult to specify minimum data requirements as this depends on the purpose for which the model is to be used. In practice, the commonest useful data will comprise information on growth and reproduction over a part of the life cycle at a constant food level. However, not all primary model parameter can be estimated from such data alone. This can already be seen from Section 2.3: we can reduce the model with ten primary parameters (Table 2.1) to a model with six secondary parameters and describe the same growth and reproduction patterns.

Estimating the two primary parameters related to feeding (F_m^a and y_{AX}) requires data on actual ingestion rates at several quantifiable food densities (X). In most cases, this is not feasible, and the scaled functional response f is used as a primary parameter (see Section 2.3). Note that under *ad libitum* feeding conditions, we can fix $f = 1$. The remaining three yield coefficients (y_{BA} , y_{VA} , y_{AV}) also cannot be independently estimated without other types of data to provide more detail on the mass balance (ingestion, defecation, respiration, shrinking on starvation, etc.), so they will usually have to remain fixed at the suggested values.

If reproduction is measured as number of offspring, we need to estimate the dry weight per egg W_{B0} , which can be measured directly (or estimated from egg volume). If body size is determined as a physical length measure, we need to estimate a shape corrector δ_M , and the dry-weight density of tissue d_V (see Eq. 2.17). If body size is determined as wet weight, we only need the latter conversion factor. This leaves four primary parameters to be estimated by optimisation to the growth and reproduction data: J_{Am}^a , J_M^v , κ and W_{Vp} .

The case study in the Chapter 3 demonstrates in more detail how parameters can

be estimated from commonly available data on life history. Even though the case study comprises three food levels, it should be stressed that one level (preferably *ad libitum*) would have sufficed.

2.6 Model summary

Model assumptions are summarised in Table 2.2, and the model equations in Table 2.3. Model parameters already have been collected in Table 2.1.

-
1. There are three types of biomass: food, assimilates and structural body components. Each type has a constant composition. They can be converted into each other with a certain constant efficiency. The state variables of the organism are the masses of the structural body, the reproduction buffer for adults, and the egg buffer used by the developing embryo. Total body mass is the sum of structure and reproduction buffer in adults, and the sum of structure and egg buffer for eggs. The reproduction and egg buffer consist of assimilates.
 2. The animal has three life stages: an embryo that does not feed but utilises the egg buffer, a juvenile that feeds but does not reproduce, and an adult that reproduces. The embryo starts with an egg buffer of assimilates and negligible structural mass. The first transition (birth) is triggered by the depletion of the egg buffer, and the second transition (puberty) by reaching a critical structural body weight.
 3. The maximum feeding rate is proportional to the surface area of the animal. The animal is either searching for food or handling it (with constant handling time), leading to a hyperbolic functional response in the food density (Holling type II).
 4. Food is instantly translated into assimilates that are directly used to fuel metabolic processes. Embryos assimilate their egg buffer at the maximum rate for their structural size.
 5. The flow of assimilates is split into a constant fraction κ for maintenance and structural growth (the soma), and $1 - \kappa$ for maturation and reproduction. From the κ flow, maintenance costs are paid first. Only structural biomass requires maintenance, which is proportional to its volume. The remainder of this flow is used for growth (with certain efficiency).
 6. For adults, the $1 - \kappa$ flow is used to fill the reproduction buffer. For embryos and juveniles, all of the assimilates in this flux are burnt to increase complexity of the organism. At spawning events, the contents of the reproduction buffer are converted into eggs. The part of the buffer that was insufficient to create a single egg remains in the buffer. Transformation of buffer to egg comes with a certain (generally high) efficiency.
 7. If feeding is insufficient to pay somatic maintenance costs, the organism first diverts energy from the $1 - \kappa$ flux of assimilates and from the reproduction buffer. If that is insufficient, structure is converted into assimilates to pay maintenance.
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Table 2.2: The list of assumptions that leads to the basic DEBkiss model.

2.7 Software

A BYOM package for Matlab is available for the basic DEBkiss model at <http://www.debtox.info/byom.html>. This is the package called DEBkiss, which contains a number of formulations. There is a formulation using W_{Vp} as input parameter, but also one with physical length at puberty L_p^M (which is mathematically equivalent but may be more practical in some cases). Both versions have specific sub-folders for analyses with and without the embryo stage. Maturity maintenance (see Section 4.1) can be included

Specification of feeding	
Maximum feeding	$J_{Xm}^a = J_{Am}^a / y_{AX}$
Half-saturation	$K = J_{Xm}^a / F_m^a$
Functional response	$f = \frac{X}{X+K}$
Body volume	$L^3 = W_V / d_V$
Feeding	$J_X = f J_{Xm}^a L^2$ (if $W_B > 0$ then $J_X = 0$)
Specification of assimilate fluxes	
Assimilation	$J_A = f J_{Am}^a L^2$ (if $W_B > 0$ then $f = 1$)
Somatic maintenance	$J_M = J_M^v L^3$
Structural growth	$J_V = y_{VA}(\kappa J_A - J_M)$
Reproduction buffer	$J_R = (1 - \kappa)J_A$ (if $W_V < W_{Vp}$ then $J_R = 0$)
Assimilate fluxes under starvation ($\kappa J_A < J_M$)	
$J_A > J_M$ or $W_R > 0$	$J_V = 0$ and $J_R = J_A - J_M$ (if $W_V < W_{Vp}$ then $J_R = 0$)
$J_A \leq J_M$ and $W_R = 0$	$J_V = (J_A - J_M) / y_{AV}$ and $J_R = 0$
State variables	
Egg buffer	$\frac{d}{dt} W_B = -J_A$ with $W_B(0) = W_{B0}$ (until $W_B = 0$)
Structural body mass	$\frac{d}{dt} W_V = J_V$ with $W_V(0) \approx 0$
Reproduction buffer	$\frac{d}{dt} W_R = J_R$ with $W_R(0) = 0$
Spawning events	
Number of eggs	$\Delta R = \text{floor}(y_{BA} W_R / W_{B0})$
Reset buffer	$W_R = W_R - \Delta R W_{B0} / y_{BA}$
Continuous reproduction	$R = y_{BA} J_R / W_{B0}$
Derived model results	
Max. structural mass	$W_{Vm} = d_V (\kappa J_{Am}^a / J_M^v)^3$
Growth rate constant	$r_B = y_{VA} J_M^v / (3d_V)$
Max. reproduction rate	$R_m = (1 - \kappa) J_{Am}^a L^2 y_{BA} / W_{B0}$
Mass at birth	$W_{Vb} = W_{B0} y_{VA} \kappa$ (when $J_M \approx 0$)
Hatching time	$t_b = 3W_{B0}^{1/3} d_V^{2/3} / (J_{Am}^a (y_{VA} \kappa)^{2/3})$ (when $J_M \approx 0$)
Physical length	$L^{\mathcal{M}} = L / \delta_{\mathcal{M}}$
Total weight	$W_W = W_V + W_R$

Table 2.3: Summary of the complete model definition and some derived model results. The ‘floor’ function for the spawning events means rounding to the nearest integer less than the value between brackets.

or excluded by setting a simple option. Additionally, the package includes a basic model formulation in compound parameters (including maturity maintenance).